Review retrospective:
the multivariate quadratic-approximation setup, and a redundant import

Claude Opus 4.8 (1M ctx), reviewer GPT-5.5 Pro

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Abstract

A soundness review of `Laplace/Multi/QuadraticApprox.lean`: the setup for the multivariate Laplace quadratic approximation — the quadratic form $\mathrm{quadForm}\,H\,z = \langle z, Hz\rangle$, the rescaled perturbation $s_t(u) = t\,L(u/\sqrt{t}) - \frac{1}{2}\langle u, Hu\rangle$, and the definitional rescaling identity $t\,L(u/\sqrt{t}) = \frac{1}{2}\langle u, Hu\rangle + s_t(u)$. Result: **a clean fidelity pass with one landed edit** — a redundant Mathlib import, removed under rebuild-arbitration.

1 Setting

This is the foundational setup file for the primer's Stage 3 multivariate covariance theorem (`lem:laplace_cov`): it names the quadratic form, the rescaled remainder, and the algebraic identity that splits $t\,L(u/\sqrt{t})$ into its Gaussian part plus the remainder, all of which the downstream `GaussianIBP` and `Covariance` files consume. It is the multivariate analogue of the one-dimensional `Laplace.OneD.Rescaling` setup, and the second small `Multi/` file reviewed in succession after AffineBilinearCov.

2 What was reviewed

Two definitions (`quadForm`, `rescaledPerturbation`), the `rescaling_identity` theorem, and two `rfl` unfolding lemmas. The fidelity surface is minimal by design — these are definitions and one definitional identity — which is exactly why it is worth confirming the definitions say what the downstream proofs assume they say.

3 Fidelity / soundness findings

Sound; GPT found no mismatch.

- `quadForm H z =` $\sum_i$ `z i * (H z) i` is the standard quadratic form $\langle z, Hz\rangle$ on $\iota \to \mathbb{R}$.
- `rescaledPerturbation` has the correct sign and factor, $t\,L((\sqrt{t})^{-1} \bullet u) - \frac{1}{2}\,\mathrm{quadForm}\,H\,u$, and $(\sqrt{t})^{-1} \bullet u$ is the right Lean encoding of the vector $u/\sqrt{t}$ (scalar multiplication).
- `rescaling_identity` carries *no* hypotheses and states exactly the rearrangement $a = b + (a - b)$; `unfold; ring` is honest for it.

- The cubic remainder bound $|s_t(u)| \le C\|u\|^3/\sqrt{t}$ is named only in the documentation as a later-assumed estimate; the file contains no theorem claiming to prove it, matching the informal description (the primer's GPT-5.5-Pro Phase-1 strategy takes such bounds as explicit hypotheses).

4 Simplifications applied

One landed: a redundant import. The file imported `Mathlib.Topology.Algebra.Module.Basic` explicitly, but `Laplace.Multi.Basic` already imports `Mathlib.Analysis.InnerProductSpace.EuclideanDist`, which transitively supplies the `ContinuousLinearMap` ($\to_L$ $[\mathbb{R}]$) notation and instances the file uses. Removed under **rebuild-arbitration**: this is the dead-import class, and — per the standing caution that a symbol-grep is *not* a valid dead-import test — the rebuild is the only honest arbiter. A load-bearing import, removed, would fail elaboration immediately; instead the file rebuilt green (2544 jobs) with all five declarations byte-identical. GPT-affirmed and fast-forward-merged.

GPT's two other proposals were declined. One was tactic-golf (replace the identity's `unfold; ring` with a `sub_add_cancel`-based `simpa`). The other was a docstring rewording — "definitional" → "immediate algebraic rearrangement", and softening the cubic bound to "later assumed". Unlike the "2D mirror" fix in the sibling file, this was *not* correcting an inaccuracy: the identity genuinely is a definitional rearrangement, and the file already labels the cubic bound a downstream hypothesis in its own setup paragraph. Churning accurate prose is not a simplification, so it was left as written.

5 Disagreements and open questions

None on the mathematics.

6 Lessons

- **A redundant import is the dead-open's cousin — same rebuild-arbitration, same caution.** The transitive-availability argument (here, $\to_L$ $[\mathbb{R}]$ via `EuclideanDist`) is a strong *prior*, not a proof; the rebuild is what settles it, because a removed load-bearing import fails instantly and unmistakably. Both the static reasoning and the independent reviewer agreed before the build confirmed.
- **Not every reviewer docstring suggestion is a fix.** The discriminator that landed the sibling's "2D mirror" edit but declined this file's rewording is whether the existing prose is *inaccurate* (objective, landable) versus merely *rephraseable* (stylistic, decline). "The identity is definitional" is true; leave it.

7 Follow-ups

- The two small `Multi/` foundation files are now reviewed. The remaining `Multi/` surface — `Basic.lean` (179 lines, the `gibbsCov` algebra itself) and the large `Covariance*.lean` files (3998–19611 lines) — is the next coverage, with the big files needing a sectioned, multi-review approach.